Optimizing Prompts for Extracting Bitcoin Price Predictions from News Articles

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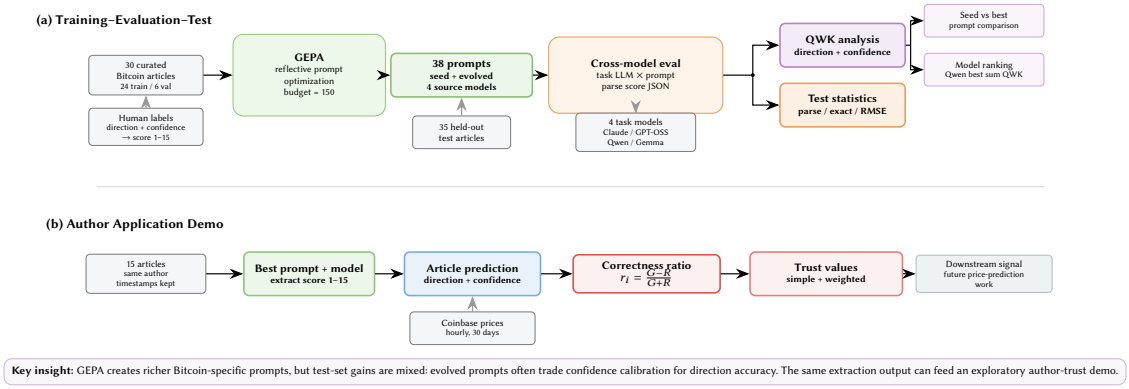

Figure 1: Overview of the Bitcoin news prompt optimization study.

Abstract

This study asks whether prompt optimization can help large language models extract structured Bitcoin price predictions from news articles. We use GEPA to optimize prompts on 30 manually labeled articles, then evaluate 38 seed and evolved prompt candidates across four task models on a 35-article held-out test set. Predictions are mapped to a 1–15 ordinal score that combines direction and confidence, and are evaluated with parse statistics, exact and direction accuracy, RMSE, and Quadratic Weighted Kappa (QWK). GEPA produced longer, more domain-specific prompts, but the selected best prompts did not consistently outperform the shared seed prompt on test QWK. Across task models, Claude Sonnet 4.6 achieved the strongest direction agreement, while Qwen 3.6 achieved the highest combined direction and confidence QWK. These results suggest that optimized prompting can structure financial news sentiment, but larger datasets and direction-aware training metrics are needed for reliable downstream use.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Machine learning*; • **Information systems** → *Sentiment analysis.*

Keywords

prompt optimization, GEPA, financial sentiment, LLM evaluation

1 Introduction

1.1 Motivation

Bitcoin (BTC) price coverage is a very nuanced space. The market was born into the digital age, where many traders take to either social media or personal blogs in order to share their sentiments in very informal ways. Many of these predictions take on the form of messy natural language, rather than more traditional finance information tidbits. A single article or post in the BTC space may combine things like analyst quotes, support and resistance levels, macroeconomic contexts, and hedging language. This makes it uniquely difficult to take signals in from the market and make a prediction as to whether an article believes a price is to go up or down in the near future. Furthermore, BTC is one of the most volatile assets on the planet, with sentiment updates and trading occurring around the clock. We want to compare what an article is actually saying versus what later happened in the market.

We were motivated by the desire to turn the signals written into the casual sentiment space into structured data. Instead of asking a model to predict BTC's price directly, we ask the question of whether an optimized prompt can extract two features from an article published into the space: direction and confidence. Does the author predict a price surge or a tailspin? How confident do they seem in that prediction, and are they hedging their bets? We believe this mechanism creates something more distinctive than a traditional trading algorithm. Instead, we set out to create a careful reader that operates on how the market feels rather than simply how it looks.

Prompt optimization became the clear path because our dataset is small and had to be manually labeled. With such a high cost of harvesting, we skipped model fine-tuning and used GEPA instead. This allows our system to search for a better set of instructions that make LLMs more efficient at producing something as close to the human gold standard as possible. If this approach works, it will suggest that optimized prompting is an efficient way of converting unstructured financial news into analyzable signals. This can later be compared across models, authors, and realized BTC price movements.

1.2 Research Questions

Q1: Does GEPA produce prompts that improve structured extraction of forward-looking Bitcoin sentiment compared with the shared seed prompt?
Q2: Which task model best recovers human-labeled direction and confidence on the held-out Bitcoin news test set?

1.3 Pipeline Overview

As shown in Figure 1, our pipeline first uses GEPA to optimize prompts on a small manually labeled training set of Bitcoin prediction articles. We then evaluate all generated prompt candidates across multiple LLMs on a held-out test set, measuring how closely each model-prompt pair recovers human gold sentiment scores. Finally, we use the strongest prompt-model configuration for an optional author-level validation analysis comparing extracted article predictions with later Bitcoin price movement.

2 Related Work

2.1 Prompt Optimization

Prompt optimization improves model behavior by modifying instructions rather than updating model weights. Compared to manual prompt engineering, methods such as DSPy [5], OPRO [9], and GEPA [1] provide a systematic way to search over prompt variants. GEPA (Genetic-Pareto) frames this as reflective prompt evolution: candidate prompts are generated, evaluated on a task metric, and iteratively refined using feedback from the model's reasoning, while the underlying LLM remains fixed and is not fine-tuned. This makes GEPA particularly useful in settings where the goal is to extract reliable structured outputs without additional model training.

2.2 LLM Financial Sentiment Analysis

Financial sentiment analysis uses natural language processing to classify and extract market-relevant signals from an article or post. This is harder than general sentiment analysis as there is market-specific language that is specialized. For example, hedging and bullish vocabulary could mean something completely different out of context. Models such as FinBERT show the value of fine-tuning models for financial text. Their developers show that general sentiment analysis models struggle with domain-specific phrases, along with limited labeled data [3]. More recent work applies larger models to financial documents. Newer LLMs can perform richer extraction tasks that go above and beyond simple binary classification [8]. Cryptocurrency sentiment analysis is even more difficult than standard asset price predictions, as it trades continuously. Bitcoin also reacts nearly instantaneously to new information, regardless of the time, as there are no markets that are required to be open to clear trades. Our work aims to build on this literature, but shifts the target from simple predictions to extracting the direction and confidence of predictions embedded in articles.

3 Method

3.1 Models

We evaluate four Large Language Models (LLMs): Claude Sonnet 4.6, GPT-OSS-120B, Qwen 3.6, and Gemma 4 (E2B variant). These models were chosen because they represent the current state of the art across different model families, offering a wide range of parameter scales and architectural designs. We used an API key to access Claude and pulled the other three models from Ollama. LiteLLM was used to call all models through a unified interface that standardizes interactions with different LLM providers through a consistent API.

3.1.1 Claude Sonnet 4.6 (Anthropic). This proprietary model is based on a transformer architecture. It incorporates reinforcement learning from human feedback for alignment and supports near-instant responses or extended step-by-step thinking, with a 1M-token API context window in beta [2].

3.1.2 GPT-OSS-120B. This is a transformer mixture-of-experts model from OpenAI with 117 billion total parameters and a 128K context length, trained on large-scale web and code data. It is open-weight and designed for local inference [7].

3.1.3 Qwen3.6:35b. This model from Alibaba is trained on multilingual and code-heavy datasets with a 256K context window. Its distinguishing feature is strong multilingual capability combined with competitive reasoning performance [6].

3.1.4 Gemma 4 (E2B). The E2B variant of this open-weight model developed by Google is a relatively smaller-scale model compared to frontier systems, designed for efficient local deployment while maintaining reasonable performance. It is trained on a mixture of web data, code, and curated datasets, with an emphasis on lightweight reasoning and general language tasks [4].

3.2 Data

The dataset was curated through manual selection from online sources, specifically filtering for content containing predictive insights rather than retrospective summaries. This ensures the model learns to identify forward-looking signals rather than merely processing historical narratives.

To establish a human gold standard, each entry was manually labeled based on two dimensions: direction and confidence. Direction classifies the nature of the predicted outcome using a categorical scale of -1, 0, and $+1$, while confidence represents the certainty expressed by the author on a scale of 1 to 5. These qualitative assessments were then mapped onto a unified 1–15 numerical score, serving as the primary ground truth for training and evaluating the model's predictive accuracy. Additionally, each entry includes human-annotated reasoning notes that explain the logic behind the assigned scores. These notes provide critical context for resolving labeling ambiguities and serve as a qualitative benchmark for verifying the model's interpretability during the inference phase.

The dataset is partitioned into three segments for training, evaluation, and practical application. We used 30 labeled articles for the development phase, splitting them into 24 for training and 6 for validation, followed by a separate set of 35 articles for final testing. For the application phase, we curated 15 unlabeled articles from a single author; these include high-precision timestamps, which are essential for the longitudinal analysis discussed later in Section 3.7.

Table 1: Mapping from direction and confidence to gold score

Score	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Direction	−1	−1	−1	−1	−1	0	0	0	0	0	+1	+1	+1	+1	+1
Confidence	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5

For example, a strongly bearish article is labeled direction −1 with confidence 5, which maps to score 5. A moderately bullish article is labeled direction +1 with confidence 3, which maps to score 13.

Table 2: Notation for the GEPA optimization framework

Symbol	Meaning
Φ	Language-model system
Π	Collection of prompts
Θ	Model weights
x	Task input
m	Evaluator metadata
T	Task distribution
μ	Evaluation metric
arg max	Selects the best-scoring parameters
$\mathbb{E}$	Expected or average performance

Table 3: Partial-credit evaluator used during GEPA

Prediction outcome	Score
Exact match	1.00
Off by 1	0.75
Off by 2	0.50
Off by 3	0.25
Off by more than 3	0.00
Parse failure	0.00

3.3 Stage 1 — GEPA Prompt Optimization

The general optimization objective can be written as:

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg\max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x,m)\sim T} \left[\mu \left(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m \right) \right]. \quad (1)$$

This objective searches for prompts that maximize expected task quality. The term $\mathbb{E}_{(x,m)\sim T}$ averages performance over task instances sampled from distribution T. The operator arg max selects the parameter setting with the highest expected metric score.

In our implementation, the optimizer is called through `gepa.optimize`. For each run, the same source model is used as both the task model and the reflection model. We initialize the search with a seed prompt that asks the model to act as a Bitcoin sentiment analyst, consider only forward-looking claims, ignore retrospective price information, and return only `{"score": <integer 1-15>}`. For each article, the evaluator metadata m contains the human gold score and gold reasoning. A rollout applies one prompt candidate to task examples and compares the model's JSON score with this metadata.

The metric μ is the ordinal partial-credit score used by our custom evaluator. These values become the numeric evaluation signal that GEPA uses to compare prompt candidates.

The natural-language feedback reports whether the prediction was correct, too bullish, too bearish, or failed to parse, and it includes the human gold reasoning as context for reflection. GEPA uses this feedback to revise the prompt pool by mutating, merging, and selecting candidate prompts according to evaluated quality.

We set the optimization budget to 150 metric calls for each source-model run. This budget limits the total number of evaluator calls GEPA can spend across rollouts, so each iteration must trade off evaluating existing candidates against exploring revised prompts. We chose this budget as a practical balance: large enough to allow multiple rounds of prompt evolution, but small enough to keep the four source-model runs computationally manageable.

This process is executed separately for each of the four source models on the training dataset, using 24 articles for training and 6 articles for validation. Across these runs, GEPA produces 38 experimental prompts in total, including the validation-best prompt selected for each model.

3.4 Stage 2 — Cross-Model Test Evaluation

In Stage 2, every prompt candidate produced in Stage 1 is evaluated with every task model on the held-out test set of 35 articles. This creates a full cross-product of 4 task models, 38 prompts, and 35 test articles. For each prediction, we record the model's parsed score, the gold score, the absolute and signed errors, parse status, and the ordinal partial-credit score.

3.5 Stage 3 — QWK Agreement Analysis

The 1–15 score should not be treated as fifteen unrelated classes because it combines two ordinal judgments: direction and confidence. We therefore evaluate agreement using Quadratic Weighted Kappa (QWK), which accounts for chance agreement and penalizes larger ordinal disagreements more heavily than smaller ones. Direction and confidence are evaluated separately, allowing us to distinguish whether a model fails on the bearish/neutral/bullish boundary or only on the strength of the prediction.

Let O_{ij} be the observed confusion matrix between gold label i and predicted label j:

$$O_{ij} = \#\{n : y_n = i, \hat{y}_n = j\}. \quad (2)$$

The expected matrix under chance agreement is computed from the row and column marginals:

$$E_{ij} = \frac{r_i c_j}{N}, \quad (3)$$

where r_i is the number of gold labels in class i, c_j is the number of predictions in class j, and N is the total number of parsed predictions. For K ordinal classes, QWK uses the quadratic weight

$$w_{ij} = \frac{(i-j)^2}{(K-1)^2}. \quad (4)$$

The final weighted kappa is then

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}. \quad (5)$$

A value of 1 indicates perfect agreement, while 0 indicates chance-level agreement.

In our case, each 1–15 score is decomposed into direction and confidence: scores 1–5 are bearish with confidence 1–5, scores 6–10 are neutral with confidence 1–5, and scores 11–15 are bullish with confidence 1–5. We then compute QWK once on the three-class direction label and once on the five-class confidence label. The first analysis compares each GEPA-selected prompt against its original seed prompt to measure whether prompt optimization improves agreement. The second analysis aggregates predictions by task model to compare which model performs best on the rating task.

Condition	Label	Classification
$P_t > 1.05 \times P_0$	Bullish Escape	Red
$b \leq P_t \leq 1.05 \times P_0$	Neutral	Green
$0.95 \times P_0 \leq P_t < b$	Neutral	Green
$P_t < 0.95 \times P_0$	Bearish Escape	Red

Table 4: Price Zones for a Neutral Prediction

We compare direction QWK and confidence QWK separately and sum them to obtain the final agreement score.

3.6 Author Trustworthiness Evaluation

To calculate the reliability of an author, we implemented a scoring method that compares the authors market sentiment in an article with Bitcoin movement in the 30 day period after publication. Unlike a binary success/fail evaluation, our method captures both the magnitude and the duration price movements and rewards predictions that is reflected with overall price trends over fluctuations.

3.6.1 Bitcoin Price Data Collection. For each article, we record the Bitcoin prices hourly after publication for a month utilizing Coinbase API. The hourly granularity was a limitation of the free tier of the API. The 30-day evaluation window allows sufficient time for the author's prediction to actualize. The window may appear too short in the context of securities at large but it is an overestimate in the context of Bitcoin due to its volatility. An overestimate was done to ensure sufficient data collection.

3.6.2 Sentiment Score. 15 articles from the same bitcoin author, Manish Chhetri, was fed through Qwen 3.6 model prompted an evolved system prompt from GEPA. The scoring system worked via the following formula:

$$s_i = \Delta + c_i \tag{6}$$

where Δ is a direction-specific offset:

$$\Delta = \begin{cases} 0 & \text{if bearish} \\ 5 & \text{if neutral} \\ 10 & \text{if bullish} \end{cases} \tag{7}$$

and $c_i \in [1, 5]$ is the author's expressed confidence in their prediction. This gives three non-overlapping ranges:

- Bearish: $s_i \in [1, 5]$
- Neutral: $s_i \in [6, 10]$
- Bullish: $s_i \in [11, 15]$

3.6.3 Correctness Ratio Classification. For each hour $(t, t+1)$, we compute the segmented area relative to the baseline b (price at the time of publication) using the trapezoidal rule iteratively updating the calculation. The segment's midpoint is used to assign each as green (correct) or red (incorrect). A simple linear connection between the two price points is a notable limitation of the paper.

For bullish prediction the prediction is correct when the price of bitcoin is above the baseline and incorrect when below. For bear prediction the prediction is correct when the price of bitcoin is below the baseline and incorrect when above.

For neutral predictions, the calculation defines success as price movement within ±5% of the bitcoin price at the time of publication, which is a generous window chosen due to bitcoin's volatility.

Condition	Interpretation
$\tau > +0.5$	Consistently Correct
$-0.5 \leq \tau \leq +0.5$	Unreliable or Noisy
$\tau < -0.5$	Consistently wrong

Table 5: Author Trustworthiness Thresholds

This partitions the price space into four zones:

3.6.4 Correctness Ratio. The per-article prediction corectness ratio is defined as:

$$r_i = \frac{G - R}{G + R} \tag{8}$$

The ratio has several notable properties. The ratio is bounded to $r \in [-1, +1]$. A value of $+1$ indicates that the price was and remained correct for the entire month. A value of -1 indicates that the price was and remained incorrect for the entire month. A positive ratio indicates that the author was more right than wrong and vice versa. The data is outputted as a CSV, allowing visualization of how the ratio evolves over the 30-day window.

3.6.5 Weighted and Unweighted Scores. Two trust scores are computed with the results.

The simple trust score is the unweighted mean ratio:

$$\tau_{\text{simple}} = \frac{1}{15} \sum_{i=1}^{15} r_i \tag{9}$$

The confidence-weighted trust score:

$$\tau_{\text{weighted}} = \frac{\sum_{i=1}^{15} r_i \cdot c_i}{\sum_{i=1}^{15} c_i} \tag{10}$$

The weighted scale is the preferred trustworthiness metric as it penalizes authors who make correct but low-confidence predictions while rewarding those who make bold predictions that prove correct. The denominator $\sum c_i \geq 15$ is always a non-zero value (a positive value) since $c_i \geq 1$.

3.6.6 Price Movement and PnL Analysis. To further validate the data, we also conducted conventional analyses.

The aggregate Bitcoin price movement is calculated for each prediction over the 30-day window. This assesses whether the LLM-extracted sentiment correlates to a price movement in the predicted direction. This serves as a crude check on the scoring system.

Profit and Loss (PnL) is also calculated. It charts hypothetical profits and losses in percentage if one bought/shorted bitcoin at `t = 0` and exited the position at `t = n`. Neutral articles are excluded from the PnL calculation. Both simple and confidence-weighted aggregate PnL are reported with ±1 standard deviations.

A simple timeline is charted to shows the Bitcoin price — from October 2024 to April 2026 — with each article's publication date marked. The dot color indicates the predicted direction and the dot size indicates the confidence level. This serves to contextualize the data.

4 Results

4.1 GEPA Prompt Evolution

4.1.1 Stage 1 Basic Run Results. Stage 1 optimized prompts separately for the four source models used in the study: Claude Sonnet

Source model	Cand.	Iter.	Acc.	Best words	Growth
Claude Sonnet 4.6	11	15	10	615	6.68×
GPT-OSS 120B	7	20	6	92	1.00×
Qwen 3.6	11	14	10	714	7.76×
Gemma 4 E2B	9	16	8	259	2.82×
Total	**38**	**65**	**34**	–	–

Table 6: Basic Stage 1 GEPA results. "Best words" is the word count of the GEPA-selected best prompt for that source model, measured against the 92-word seed prompt.

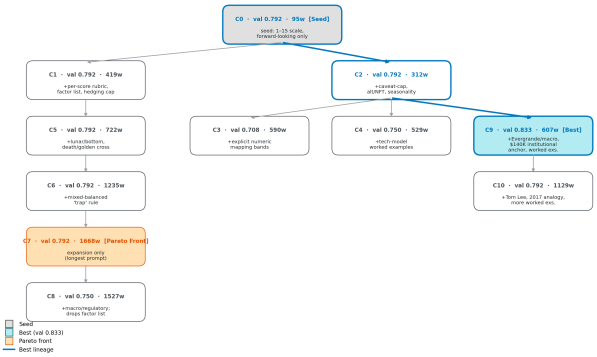

Figure 2: Example GEPA prompt evolution tree for the Claude Sonnet 4.6 run. Note: C0 is the seed prompt, and the selected best prompt is C9. The best path is C0 → C2 → C9, illustrating that GEPA can preserve multiple candidate branches while selecting a later improved prompt.

4.6, GPT-OSS 120B, Qwen 3.6, and Gemma 4 E2B. Each run started from the same 92-word seed prompt and used a budget of 150 evaluator calls on the 30-article training set. Across the four completed runs, GEPA retained 38 prompt candidates in the final candidate pools. This total consists of the four initial seed prompts plus 34 accepted evolved prompts over 65 logged optimization iterations.

Note: GPT-OSS 120B's GEPA-selected best prompt is the original seed prompt, which is why its word-count growth is 1.00×.

The main change in prompt length was expansion. Three of the four selected best prompts became substantially longer than the seed: Claude's selected prompt grew from 92 to 615 words, Qwen's from 92 to 714 words, and Gemma's from 92 to 259 words. GPT-OSS was the exception: although its run generated several longer prompt candidates, the selected best prompt remained the original 92-word seed. Overall, the Stage 1 results show that GEPA usually preferred more detailed instructions for this extraction task, but that longer prompts were not automatically selected in every model-specific run.

4.1.2 Example Prompt Evolution Tree. Figure 2 shows an example GEPA evolution tree from the Claude Sonnet 4.6 source-model run. The tree visualizes how accepted prompt candidates branch from earlier prompts rather than following a single linear revision path.

4.2 Test Evaluation Statistics

Stage 2 evaluates the prompt candidates produced by GEPA on the held-out 35-article test set. The available test-evaluation artifact contains 4,480 per-article predictions. Claude Sonnet 4.6, GPT-OSS 120B, and Gemma 4 E2B each have 1,330 predictions, corresponding to 38 prompt candidates evaluated across 35 test articles. The Qwen 3.6 portion contains 490 completed predictions in the current artifact.

Task model	Pred.	Parse	Exact	Dir.	RMSE
Claude Sonnet 4.6	1330	81.3%	13.0%	65.5%	2.17
GPT-OSS 120B	1330	99.6%	22.3%	77.0%	2.85
Qwen 3.6	805	82.2%	24.6%	76.0%	2.82
Gemma 4 E2B	1330	100.0%	23.6%	67.2%	3.57

Table 7: Stage 2 descriptive statistics by task model. Exact accuracy and direction accuracy are computed over all recorded predictions, so parse failures count as incorrect. RMSE is computed only over successfully parsed predictions.

For each task model, we report the parse success rate, exact score accuracy, direction accuracy, and root mean squared error (RMSE). Parse success measures whether the model returned a valid structured score. Exact accuracy measures whether the predicted 1–15 score exactly equals the gold score. Direction accuracy is less strict: it only checks whether the prediction falls in the same bearish, neutral, or bullish score band as the gold label. RMSE measures the typical size of numeric score error after squaring larger mistakes more heavily. In this setting, it is computed on successfully parsed predictions as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_{\text{parsed}}} \sum_{i=1}^{N_{\text{parsed}}} (\hat{y}_i - y_i)^2}, \tag{11}$$

where $\hat{y}_i$ is the predicted 1–15 score, y_i is the human gold score, and N_{parsed} is the number of predictions with a valid parsed score.

The strongest parse reliability appears in the local open-weight models: Gemma 4 E2B parses every recorded response, and GPT-OSS 120B parses nearly all responses. GPT-OSS 120B has the highest direction accuracy among the completed 1,330-prediction model runs, while Gemma 4 E2B has the largest RMSE despite its perfect parse rate. This suggests that valid formatting and numeric closeness are separate behaviors: a model can reliably produce parseable JSON while still making larger ordinal score errors.

4.3 QWK Agreement Results

4.3.1 Seed Prompt vs. GEPA-Best Prompt. The first QWK analysis compares each source model's seed prompt with its GEPA-selected best prompt under each task model. This creates $4 \times 4 = 16$ task/source comparisons. For each comparison, QWK is computed between the human gold labels and the model predictions, once for direction and once for confidence. We also report the sum of the direction and confidence QWK scores as a combined agreement score.

All source-model runs start from the same seed prompt. Therefore, differences among seed rows should not be interpreted as differences among distinct seed prompts. They reflect repeated model calls, retry behavior, parse filtering, and other run-level variation in the evaluation artifact. Even with this noise, the table still provides a general comparison between the shared seed prompt baseline and each source model's GEPA-selected best prompt.

The seed-vs-best comparison shows that GEPA optimization does not uniformly increase QWK agreement on the held-out set. Among

Task	Source	Seed QWK			Best QWK			Δ Best − Seed		
		Dir.	Conf.	Sum	Dir.	Conf.	Sum	Dir.	Conf.	Sum
Claude	Claude	0.908	0.254	1.162	0.912	0.049	0.961	0.003	-0.204	-0.201
Claude	Gemma	**0.926**	0.180	1.106	0.872	0.203	1.075	-0.054	0.023	-0.031
Claude	GPT-OSS	0.907	0.199	1.106	0.907	0.199	1.106	0.000	0.000	0.000
Claude	Qwen	**0.926**	0.135	1.061	0.892	0.211	1.103	-0.033	0.075	0.042
Gemma	Claude	0.702	0.480	1.182	0.710	0.388	1.099	0.008	-0.091	-0.083
Gemma	Gemma	0.789	0.329	1.117	0.792	0.453	1.245	0.004	0.124	0.128
Gemma	GPT-OSS	0.702	0.480	1.182	0.702	0.480	1.182	0.000	0.000	0.000
Gemma	Qwen	0.716	0.554	1.271	0.621	0.401	1.022	-0.095	-0.153	-0.248
GPT-OSS	Claude	0.845	0.561	1.407	0.875	0.345	1.220	0.030	-0.216	-0.187
GPT-OSS	Gemma	0.774	0.580	1.353	0.816	0.313	1.129	0.043	-0.267	-0.224
GPT-OSS	GPT-OSS	0.824	0.603	1.427	0.824	0.603	1.427	0.000	0.000	0.000
GPT-OSS	Qwen	0.824	0.579	1.403	0.912	0.238	1.150	0.088	-0.341	-0.252
Qwen	Claude	0.801	0.515	1.316	0.854	0.321	1.175	0.053	-0.194	-0.141
Qwen	Gemma	–	–	–	–	–	–	–	–	–
Qwen	GPT-OSS	0.814	**0.620**	**1.433**	0.814	**0.620**	**1.433**	0.000	0.000	0.000
Qwen	Qwen	0.814	0.587	1.401	–	–	–	–	–	–

Table 8: Seed-vs-best QWK comparison for all 16 task/source model pairs. Direction QWK is computed on bearish, neutral, and bullish labels; confidence QWK is computed on the 1–5 confidence label. "Sum" is the direction QWK plus confidence QWK. Dashes indicate that no parsed rows were available for that side of the comparison.

Task model	Rows	Parsed	Parse	Dir. QWK	Conf. QWK	Sum
Claude Sonnet 4.6	1330	1081	81.3%	**0.873**	0.137	1.010
Gemma 4 E2B	1330	1330	100.0%	0.636	0.338	0.974
GPT-OSS 120B	1330	1325	99.6%	0.792	0.270	1.061
Qwen 3.6	805	662	82.2%	0.794	**0.385**	**1.178**

Table 9: Overall QWK by task model, computed over all parsed predictions available for each model. Rows are all recorded predictions, Parsed is the subset used for QWK, and Sum is direction QWK plus confidence QWK.

the 14 comparisons with both seed and best scores available, the combined QWK score improves in 2 cases, is unchanged in 4 cases, and decreases in 8 cases. Several best prompts improve direction agreement while reducing confidence agreement, which suggests that the optimized prompts can become better at choosing the broad sentiment band without necessarily improving the finer confidence level inside that band.

4.3.2 Overall QWK by Task Model. The second QWK analysis aggregates all parsed Stage 2 predictions by task model, pooling across source-model prompt candidates and test articles. This view asks which task model agrees most strongly with the human labels overall, rather than whether a specific GEPA-best prompt improves over its seed. As in the first analysis, we report direction QWK, confidence QWK, and their sum.

This aggregate view gives a different picture from exact accuracy alone. Claude Sonnet 4.6 has the strongest direction agreement, but its confidence QWK is low. Qwen 3.6 has the highest combined QWK score because it pairs competitive direction agreement with the strongest confidence agreement. Gemma 4 E2B has perfect parse coverage but the lowest combined QWK, again showing that formatting reliability and label agreement are not the same behavior.

4.4 Author Evaluations

Figure 7 shows the Bitcoin price movement compared to the 15 articles with prediction and confidence. The author published predominantly bullish articles especially during the mid-2025 price peak (articles 6–9), and shifted toward bearish calls (articles 12–15) as the price tumbled. The clustering suggests the author is reacting to the market rather than predicting it.

Figure 6 shows the aggregate running ratio in the 30-day window. The simple and weighted trust scores begin positive at publication and peaks at the 12 hour mark and maintains above 0.5 for the first 72 hours. It then steadily decreased and crosses zero around day 23.

Figure 5 shows the aggregate Bitcoin price movement by predicted direction. Bearish articles show a consistently strong sustained price movement in the predicted negative direction. However, bullish and neutral predictions are much more inconsistent. In fact, the neutral price outperforms the bullish by the week 2 mark. But this can be attributed to the limited sample size of 15 articles.

Figures 3 and 4 show the simulated 30-day PnL for long and short positions respectively. Long positions on bullish articles consistently shows a negative PnL (loss), reflecting the limited bullish performance. Short positions on bearish articles, however, shows a consistently positive PnL (profit) throughout.

5 Discussion

5.1 What GEPA Changed in the Prompts

The seed-vs-best comparison suggests that GEPA did more than simply make prompts longer. The evolved prompts added more explicit financial vocabulary and more detailed scoring rules. This is visible in the domain-keyword heatmaps in Appendix Figure 8: compared with the seed prompt, evolved candidates use terms such as *bearish*, *bullish*, *neutral*, *price*, *target*, *outlook*, *analyst*, *forecast*, and *future* more often. In other words, GEPA tried to turn the original short instruction into a more domain-aware decision procedure.

However, the seed-vs-best QWK results show that this extra structure was not always beneficial on the test set. In only 2 of the 14 comparable rows did the combined QWK score improve, while 4 were unchanged and 8 decreased. The clearest pattern is a tradeoff between direction and confidence. For example, with GPT-OSS 120B as the task model and the Qwen source prompt, direction QWK increased by 0.088, but confidence QWK decreased by 0.341, lowering the summed score by 0.252. Similar tradeoffs appear for GPT-OSS with the Claude source prompt (+0.030 direction, -0.216 confidence) and Qwen with the Claude source prompt (+0.053 direction, -0.194 confidence). This suggests that GEPA often made prompts better at choosing the broad bearish/neutral/bullish band, but worse at calibrating the 1–5 confidence level inside that band. The result is that direction accuracy can improve while the overall QWK score declines.

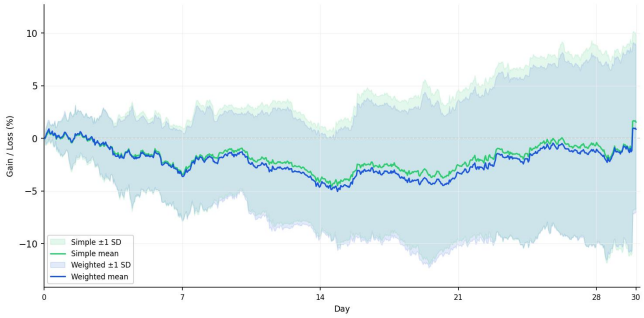

Figure 3: 30-Day Long Position PnL on Bullish Predictions

Figure 4: 30-Day Short Position PnL on Bearish Predictions

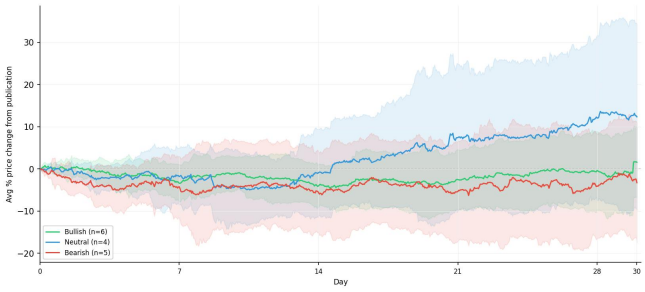

Figure 5: Bitcoin Price Movement by Predicted Direction

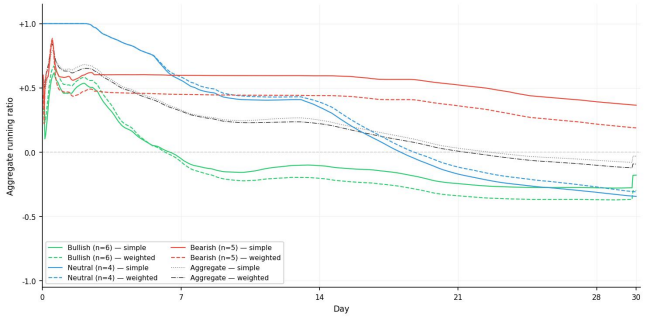

Figure 6: Trust Ratios

Figure 7: Bitcoin Price Movement Compared Against Article Publication with Confidence Scores

5.2 Model Differences in Direction and Confidence

The per-model QWK results show that models fail in different ways. Claude Sonnet 4.6 has the strongest direction QWK (0.873), which suggests that it is good at reading the broad stance of a Bitcoin article: bearish, neutral, or bullish. Its confidence QWK is much lower (0.137), meaning that after choosing the correct direction, it often struggles to place the article at the exact strength level within the 1–5 confidence scale. This pattern is consistent with Claude's design as a heavily post-trained, aligned general assistant. Anthropic describes Sonnet 4.6 as trained through substantial post-training to be helpful, honest, and harmless, with optional extended or adaptive thinking modes [2]. That design can help it make semantically safe, high-level distinctions, but it may also make the model more conservative around hedged or mixed financial language, compressing confidence scores toward the middle of a band.

Qwen 3.6 has the highest combined QWK because it has the best confidence QWK (0.385) while maintaining competitive direction QWK (0.794). This means Qwen was not the strongest model at the coarse direction boundary, but it was better at preserving ordinal distance within the score bands. A possible explanation is Qwen's explicit reasoning-oriented design. The Qwen3 technical report describes a unified model family with both thinking and non-thinking modes, plus a thinking budget for balancing latency and reasoning depth [**?**]. For our task, that structure may help the model track multiple pieces of evidence and translate them into a graded confidence score rather than only a direction label. The Qwen result therefore suggests that the best overall extraction model is not necessarily the one with the highest direction accuracy, but the one that balances direction recognition with confidence calibration.

5.3 Author-Level Findings

The data reveals a clear inconsistency: the author demonstrates meaningful skill in predicting bearish market but is limited in bullish calls. This pattern signifies a reactive author who identifies bearish signals more reliably than bullish ones, possibly because price declines tend to be faster and more obvious.

The early peak in the running aggregate ratio at 0.7 at hour 12 — maintaining above 0.5 for 72 hours — followed by sustained decline is a notable finding. It suggests that the author's predictions has some consistent signal in the immediate term — likely capturing short-term momentum that is already reflected in the article — but deteriorates over 30 days as mean-reversion and broader market forces dominate. In future research, this has implications for how an author's prediction should be interpreted ergo how automated bitcoin portfolio management should be conducted.

The ratio method completely disregards high fluctuation gains and isolates consistent gains regardless of magnitude. Therefore, in theory, the method could prove useful in the context of high-frequency trading (even outside the context of sentiment analysis).

Notably, the weighted score of the ratio is consistently lower than the simple score, indicating that the author's low-confidence prediction ratios outperforms their high-confidence ones. This statistical oddity is also observed in both long and short PnL graphs. Though the difference is only minor, it demonstrates the non-existent value of the confidence scores for this particular author. If this trend continues in other authors, future research should abandon author confidence as a metric.

5.4 Limitations

5.4.1 Limitations of the Author Evaluation. The author evaluation has several constraints. The sample size of 15 articles from a single author is insufficient for general analysis and can only serve as a proof of concept. Robust author evaluations would require a statistically significant number of articles per author across varying market conditions.

The Coinbase price data is hourly, meaning the area calculations are trapezoidal. Sharp price movements that reverse within an hour are invisible to the system. Higher granularity data would reduce the approximation error.

5.4.2 Training Metric. The GEPA training metric gives partial credit based only on absolute distance on the 1–15 score scale. This can under-penalize boundary-crossing errors: for example, predicting 11 for a gold score of 9 is only two points away, but it changes the label from neutral to bullish. As a result, the optimized prompts may learn ordinal closeness without fully respecting the bearish, neutral, and bullish direction boundaries. Although the final evaluation decomposes the score into direction and confidence for QWK analysis, this post-hoc evaluation cannot change the prompts learned during training. Future work should use a direction-aware training metric: exact match receives 1.00; same-direction predictions off by one, two, or three receive 0.75, 0.50, and 0.25 respectively; and different-direction predictions receive 0.00. Manually appending a boundary-overwrite instruction to every prompt could improve test-time outputs, but we do not apply it to the main results because it would conflate GEPA-optimized prompt behavior with post-hoc human prompt editing.

5.4.3 Parse Failures. Parse failures mostly reflect execution and formatting limitations rather than pure sentiment errors. For Claude as the task model, GPT-OSS-trained prompts often failed because they elicited step-by-step reasoning before the JSON answer. The strict parser only accepts `{"score": <integer>}` at the start of the response, so cells such as `gptoss120b__prompt_05` were recorded as `pred_score = null` even though the raw response contained substantive analysis.

Qwen 3.6 had a different failure mode. Long procedural prompts, especially GPT-OSS-source prompts, triggered very slow or empty Ollama/LiteLLM calls. The partial Qwen run had already taken about 22 hours, with many failed calls spending roughly 700 seconds in retry and timeout budget and about 705 calls still queued. We therefore killed the run early; these timed-out rows also appear as `pred_score = null`. As a result, the test-set evaluation is not perfectly aligned across all model-prompt pairs, and the missing rows limit the completeness of the experiment.

5.4.4 Dataset Size. The training and validation data are still too small for stable prompt selection. GEPA optimized prompts using only 24 training articles and 6 validation articles. On the stored validation outputs, three runs reached 5/6 exact matches (83.3%) and Claude reached 3/6 (50.0%). However, on the 35-article test set, exact accuracy was much lower: the model-level Stage 2 exact accuracies ranged from 13.0% to 24.7%. This gap suggests that some evolved prompts fit the small training/validation examples too closely and did not generalize as well to unseen articles. A larger and more diverse training set would make the GEPA selection signal less sensitive to individual articles. This overfitting risk is also an important reason why the shared seed prompt often beat the GEPA-best prompts in the seed-vs-best test comparison.

6 Conclusion

This project asked whether GEPA-optimized prompts improve structured extraction of forward-looking Bitcoin sentiment and which task model best matches human labels on the held-out test set. For the first question, the answer is mixed. GEPA successfully generated richer and more domain-specific prompts, but the GEPA-best prompts did not consistently outperform the shared seed prompt

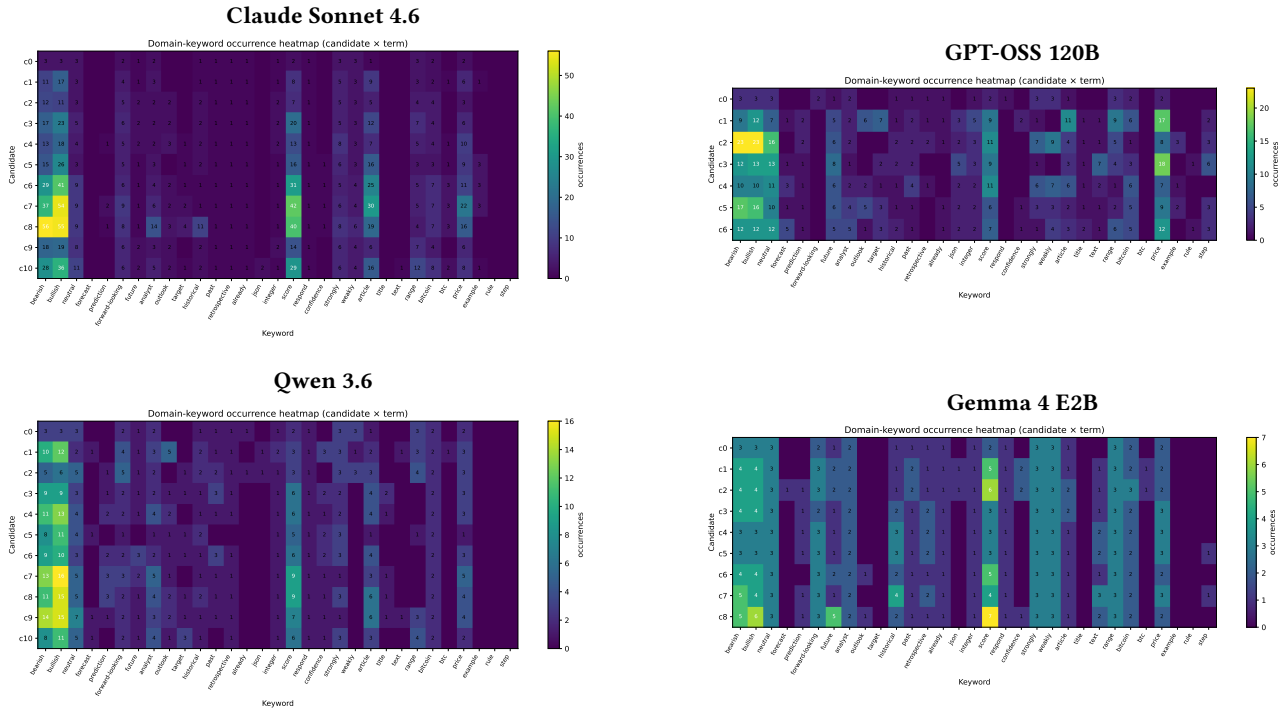

Figure 8: Domain-keyword heatmaps for the four GEPA source-model runs. Each heatmap summarizes how often Bitcoin-specific and prediction-specific vocabulary appears across the prompt candidates in that run.

on test QWK; in many cases, added prompt structure improved direction agreement while hurting confidence calibration.

For the second question, Qwen 3.6 achieved the strongest combined QWK because it had the best confidence agreement while maintaining competitive direction agreement. Claude Sonnet 4.6 was strongest on direction alone, but its weaker confidence QWK lowered its combined score. Overall, the results suggest that optimized prompting is a useful and reproducible approach for extracting prediction signals from Bitcoin news, but future work needs larger data and better training metrics before these extracted signals can reliably support price-prediction or author-trust analysis.

This study also provides a reasonable and reproducible workflow for testing whether optimized prompts can extract structured prediction information from Bitcoin news. Future work should first increase the number and diversity of labeled articles, then repeat the GEPA and test-evaluation stages on a larger split. The scoring metric should also be improved and compared across variants, especially metrics that penalize direction-boundary errors more strongly than within-band confidence errors. Finally, the author-level trust value method should be developed more deeply across more authors, longer time windows, and more market conditions, so that extracted news predictions can make a stronger contribution to actual Bitcoin price prediction.

A Domain-Keyword Heatmaps

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